

Table 55. ADC characteristics (continued)

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
f _{TRIG}	External trigger frequency	f _{ADC} = 35 MHz; 12 bits	-	-	2.33	MHz
		12 bits	-	-	f _{ADC} /15	
V _{AIN}	Conversion voltage range	-	0	-	V _{REF+} ⁽²⁾	V
R _{AIN}	External input impedance	-	-	-	50	kΩ
C _{ADC}	Internal sample and hold capacitor	-	-	5	-	pF
t _{STAB}	ADC power-up time	LDO already started	2			Conversion cycle
t _{CAL}	Calibration time	f _{ADC} = 35 MHz	2.35			μs
		-	82			1/f _{ADC}
WLATENCY	ADC_DR register write latency	CKMODE = 00	1.5/f _{ADC} + 2/f _{PCLK}	-	1.5/f _{ADC} + 3/f _{PCLK}	-
		CKMODE = 01	4.5			1/f _{PCLK}
		CKMODE = 10	8.5			
		CKMODE = 11	2.5			
t _{LATR}	Trigger conversion latency for regular and injected channels without aborting the conversion	CKMODE = 00	2	-	3	1/f _{ADC}
		CKMODE = 01	6.5			1/f _{PCLK}
		CKMODE = 10	12.5			
		CKMODE = 11	3.5			
t _s	Sampling time	f _{ADC} = 35 MHz	0.043	-	4.59	μs
			1.5	-	160.5	1/f _{ADC}
t _{ADCVREG_STUP}	ADC voltage regulator start-up time	-	-	-	20	μs
t _{CONV}	Total conversion time (including sampling time)	f _{ADC} = 35 MHz Resolution = 12 bits	0.40	-	4.95	μs
		Resolution = 12 bits	t _s + 12.5 cycles for successive approximation = 14 to 173			1/f _{ADC}
t _{IDLE}	Laps of time allowed between two conversions without rearm	-	-	-	100	μs
I _{DDA(ADC)}	ADC consumption from V _{DDA}	f _s = 2.5 MSps	-	410	-	μA
		f _s = 1 MSps	-	164	-	
		f _s = 10 kSps	-	17	-	